AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM271
Machine Learning and Deep Learning

Final Examination
Summer Semester 2025/2026

Time Allowed: 3 Hours

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Instructions:

- Total Marks: **50**
- This exam consists of **3** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are **NOT** allowed.

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PART – ONE Total Marks 10

Question No 1: Select the correct choice for each of the following statements

1. What is the main role of filters/kernels in a convolutional layer??

A) To extract important patterns from input data
B) To store the model output
C) To increase the number of classes
D) To remove the target variable

2. Which neural network is commonly used for sequential data such as time series or text?

A) CNN
B) RNN
C) K-Means
D) Decision Tree

3. In Natural Language Processing, what does tokenization mean?

A) Removing all numerical values from text
B) Splitting text into smaller units such as words or subwords
C) Converting images into pixels
D) Reducing the number of classes

4. Why is the softmax function commonly used in the output layer of a multiclass classification model?

A) To convert outputs into class probabilities
B) To reduce the size of the dataset
C) To extract image features
D) To remove noisy data

5. What is the main purpose of a recommender system?

A) To suggest suitable items to users based on their preferences
B) To compress images
C) To clean missing values only
D) To convert speech into text

PART – TWO Total Marks 15

Question No 2: Differentiate between the architectures of AlexNet and ResNet. **(6)**

Question No 3: Explain how you would build a model to predict several future values in a time series problem. **(5)**

Question No 4: Explain recommender systems and mention three real-world applications. **(4)**

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PART – THREE Total Marks 25

Scenario: Predicting Population Growth with Deep Learning

You are working as a data scientist for an energy company. The company wants to predict electricity consumption based on historical usage and weather-related factors. You are given a dataset named "*electricity_data.csv*". The dataset contains 80,000 rows and includes features such as temperature, humidity, previous electricity usage, day type, and time-related information.
Your task is to design and train a Convolutional Neural Network model to predict electricity consumption.

Question No 5: Loading the Dataset **(3)**
Write the Python code to load the dataset "population_data.csv" into a "dataset" DataFrame.

Question No 6: Dataset Splitting **(5)**
Write the Python code to split the "dataset" into training and testing datasets. Assume that there are 80,000 rows in the "dataset."

Question No 7: CNN Architecture Design **(12)**
Design a CNN architecture with 3 layers for predicting population growth. Justify the choice of layers, activation functions, and any other architectural decisions. You have the flexibility to adjust the number of layers as needed.

Question No 8: Model Training and Prediction **(5)**
Write a Python code to train the designed CNN model using the training dataset. Also, provide code for making predictions on the test dataset.

End of questions

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